Following the same procedure that led to (2.272), we can show that the third eigenvector is given by

$$| a_3\rangle = \begin{pmatrix} 0 \\ y \\ -i(1+\sqrt{2})y \end{pmatrix}; \tag{2.273}$$

its normalization leads to $y = 1/\sqrt{2(2+\sqrt{2})}$ (we have considered only the positive value of y); hence

$$| a_3\rangle = \begin{pmatrix} 0 \\ \frac{1}{\sqrt{2(2+\sqrt{2})}} \\ -\frac{i(1+\sqrt{2})}{\sqrt{2(2+\sqrt{2})}} \end{pmatrix}. \tag{2.274}$$

2.6 Representation in Continuous Bases

In this section we are going to consider the representation of state vectors, bras, and operators in *continuous* bases. After presenting the general formalism, we will consider two important applications: representations in the *position* and *momentum* spaces.

In the previous section we saw that the representations of kets, bras, and operators in a discrete basis are given by discrete matrices. We will show here that these quantities are represented in a *continuous* basis by *continuous matrices*, that is, by noncountable infinite matrices.

2.6.1 General Treatment

The orthonormality condition of the base *kets* of the continuous basis $| \chi_k\rangle$ is expressed not by the usual discrete Kronecker delta as in (2.170) but by Dirac's *continuous delta* function:

$$\langle \chi_k | \chi_{k'}\rangle = \delta(k'-k), \tag{2.275}$$

where k and k' are continuous parameters and where $\delta(k'-k)$ is the Dirac delta function (see Appendix A), which is defined by

$$\delta(x) = \frac{1}{2\pi}\int_{-\infty}^{+\infty} e^{ikx}dk. \tag{2.276}$$

As for the completeness condition of this continuous basis, it is not given by a discrete sum as in (2.172), but by an integral over the continuous variable

$$\int_{-\infty}^{+\infty} dk \; | \chi_k\rangle\langle\chi_k | = \hat{I}, \tag{2.277}$$

where $\hat{I}$ is the unit operator.

Every state vector $| \psi\rangle$ can be expanded in terms of the complete set of basis *kets* $| \chi_k\rangle$:

$$| \psi\rangle = \hat{I} | \psi\rangle = \left(\int_{-\infty}^{+\infty} dk \; | \chi_k\rangle\langle\chi_k |\right) | \psi\rangle = \int_{-\infty}^{+\infty} dk\, b(k) | \chi_k\rangle, \tag{2.278}$$

where $b(k)$, which is equal to $\langle \chi_k \mid \psi \rangle$, represents the projection of $\mid \psi \rangle$ on $\mid \chi_k \rangle$.

The norm of the discrete base *kets* is finite ($\langle \phi_n \mid \phi_n \rangle = 1$), but the norm of the continuous base *kets* is infinite; a combination of (2.275) and (2.276) leads to

$$\langle \chi_k \mid \chi_k \rangle = \delta(0) = \frac{1}{2\pi} \int_{-\infty}^{+\infty} dk \quad \longrightarrow \quad \infty. \tag{2.279}$$

This implies that the kets $\mid \chi_k \rangle$ are not square integrable and hence are not elements of the Hilbert space; recall that the space spanned by square-integrable functions is a Hilbert space. Despite the divergence of the norm of $\mid \chi_k \rangle$, the set $\mid \chi_k \rangle$ does constitute a valid basis of vectors that span the Hilbert space, since for any state vector $\mid \psi \rangle$, the scalar product $\langle \chi_k \mid \psi \rangle$ is finite.

The Dirac delta function

Before dealing with the representation of kets, bras, and operators, let us make a short detour to list some of the most important properties of the Dirac delta function (for a more detailed presentation, see Appendix A):

$$\delta(x) = 0, \qquad \text{for} \qquad x \neq 0, \tag{2.280}$$

$$\int_a^b f(x)\delta(x - x_0)\, dx = \begin{cases} f(x_0) & \text{if} \quad a < x_0 < b, \\ 0 & \text{elsewhere,} \end{cases} \tag{2.281}$$

$$\int_{-\infty}^{\infty} f(x) \frac{d^n \delta(x-a)}{dx^n}\, dx = (-1)^n \left. \frac{d^n f(x)}{dx^n} \right|_{x=a}, \tag{2.282}$$

$$\delta(\vec{r} - \vec{r}\,') = \delta(x - x')\delta(y - y')\delta(z - z') = \frac{1}{r^2 \sin\theta} \delta(r - r')\delta(\theta - \theta')\delta(\varphi - \varphi'). \tag{2.283}$$

Representation of kets, bras, and operators

The representation of kets, bras, and operators can be easily inferred from the study that was carried out in the previous section, for the case of a discrete basis. For instance, the ket $\mid \psi \rangle$ is represented by a single column matrix which has a continuous (noncountable) and infinite number of components (rows) $b(k)$:

$$\mid \psi \rangle \quad \longrightarrow \quad \begin{pmatrix} \vdots \\ \langle \chi_k \mid \psi \rangle \\ \vdots \end{pmatrix}. \tag{2.284}$$

The bra $\langle \psi \mid$ is represented by a single row matrix which has a continuous (noncountable) and infinite number of components (columns):

$$\langle \psi \mid \longrightarrow (\cdots\cdots \quad \langle \psi \mid \chi_k \rangle \quad \cdots\cdots). \tag{2.285}$$

Operators are represented by square continuous matrices whose rows and columns have continuous and infinite numbers of components:

$$\hat{A} \longrightarrow \begin{pmatrix} \ddots & \vdots & \ddots \\ \cdots & A(k, k') & \cdots \\ \ddots & \vdots & \ddots \end{pmatrix}. \tag{2.286}$$

As an application, we are going to consider the representations in the position and momentum bases.

2.6.2 Position Representation

In the position representation, the basis consists of an infinite set of vectors $\{|\,\vec{r}\rangle\}$ which are eigenkets to the position operator $\hat{\vec{R}}$:

$$\hat{\vec{R}}\,|\,\vec{r}\rangle \;=\; \vec{r}\,|\,\vec{r}\rangle, \tag{2.287}$$

where $\vec{r}$ (without a hat), the position vector, is the eigenvalue of the operator $\hat{\vec{R}}$. The orthonormality and completeness conditions are respectively given by

$$\langle\vec{r}\,|\,\vec{r}\,'\rangle = \delta(\vec{r}-\vec{r}\,') \;\;=\;\; \delta(x-x')\delta(y-y')\delta(z-z'), \tag{2.288}$$

$$\int d^3r\,|\,\vec{r}\rangle\langle\vec{r}\,| \;\;=\;\; \hat{I}, \tag{2.289}$$

since, as discussed in Appendix A, the three-dimensional delta function is given by

$$\delta(\vec{r}-\vec{r}\,') = \frac{1}{(2\pi)^3}\int d^3k\, e^{i\vec{k}\cdot(\vec{r}-\vec{r}\,')}. \tag{2.290}$$

So every state vector $|\,\psi\rangle$ can be expanded as follows:

$$|\,\psi\rangle \;=\; \int d^3r\,|\,\vec{r}\rangle\langle\vec{r}\,|\,\psi\rangle \;\equiv\; \int d^3r\;\psi(\vec{r})\,|\,\vec{r}\rangle, \tag{2.291}$$

where $\psi(\vec{r})$ denotes the components of $|\,\psi\rangle$ in the $\{|\,\vec{r}\rangle\}$ basis:

$$\langle\vec{r}\,|\,\psi\rangle \;=\; \psi(\vec{r}). \tag{2.292}$$

This is known as the *wave function* for the state vector $|\,\psi\rangle$. Recall that, according to the probabilistic interpretation of Born, the quantity $|\,\langle\vec{r}\,|\,\psi\rangle\,|^2\,d^3r$ represents the probability of finding the system in the volume element d^3r.

The scalar product between two state vectors, $|\,\psi\rangle$ and $|\,\phi\rangle$, can be expressed in this form:

$$\langle\phi\,|\,\psi\rangle \;=\; \langle\phi\,|\left(\int d^3r\;\;|\,\vec{r}\rangle\langle\vec{r}\,|\right)|\,\psi\rangle = \int d^3r\,\phi^*(\vec{r})\psi(\vec{r}). \tag{2.293}$$

Since $\hat{\vec{R}}\,|\,\vec{r}\rangle = \vec{r}\,|\,\vec{r}\rangle$ we have

$$\langle\vec{r}\,'\,|\,\hat{\vec{R}}^{\,n}\,|\,\vec{r}\rangle = \vec{r}^{\,n}\delta(\vec{r}\,'-\vec{r}). \tag{2.294}$$

Note that the operator $\hat{\vec{R}}$ is Hermitian, since

$$\begin{aligned}\langle\phi\,|\,\hat{\vec{R}}\,|\,\psi\rangle \;\;&=\;\; \int d^3r\,\vec{r}\,\langle\phi\,|\,\vec{r}\rangle\langle\vec{r}\,|\,\psi\rangle \;=\; \left[\int d^3r\,\vec{r}\,\langle\psi\,|\,\vec{r}\rangle\langle\vec{r}\,|\,\phi\rangle\right]^* \\ &=\;\; \langle\psi\,|\,\hat{\vec{R}}\,|\,\phi\rangle^*.\end{aligned} \tag{2.295}$$

2.6.3 Momentum Representation

The basis $\{\mid \vec{p}\rangle\}$ of the momentum representation is obtained from the eigenkets of the momentum operator $\hat{\vec{P}}$:

$$\hat{\vec{P}} \mid \vec{p}\rangle = \vec{p} \mid \vec{p}\rangle, \tag{2.296}$$

where $\vec{p}$ is the momentum vector. The algebra relevant to this representation can be easily inferred from the position representation. The orthonormality and completeness conditions of the momentum space basis $\mid \vec{p}\rangle$ are given by

$$\langle \vec{p} \mid \vec{p}\,'\rangle = \delta(\vec{p} - \vec{p}\,') \qquad \text{and} \qquad \int \mathrm{d}^3\mathrm{p} \mid \vec{\mathrm{p}}\rangle\langle\vec{\mathrm{p}} \mid = \hat{\mathrm{I}}. \tag{2.297}$$

Expanding $\mid \psi\rangle$ in this basis, we obtain

$$\mid \psi\rangle = \int d^3p \mid \vec{p}\rangle\langle\vec{p} \mid \psi\rangle = \int d^3p\, \Psi(\vec{p}) \mid \vec{p}\rangle, \tag{2.298}$$

where the expansion coefficient $\Psi(\vec{p})$ represents the *momentum space wave function*. The quantity $\mid \Psi(\vec{p}) \mid^2 d^3p$ is the probability of finding the system's momentum in the volume element d^3p located between $\vec{p}$ and $\vec{p} + d\vec{p}$.

By analogy with (2.293) the scalar product between two states is given in the momentum space by

$$\langle\phi \mid \psi\rangle = \langle\phi \mid \left(\int d^3p \mid \vec{p}\rangle\langle\vec{p} \mid\right) \mid \psi\rangle = \int d^3p\, \Phi^*(\vec{p})\Psi(\vec{p}). \tag{2.299}$$

Since $\hat{\vec{P}} \mid \vec{p}\rangle = \vec{p} \mid \vec{p}\rangle$ we have

$$\langle\vec{p}\,' \mid \hat{\vec{P}}^{\,n} \mid \vec{p}\rangle = \vec{p}^{\,n}\delta(\vec{p}\,' - \vec{p}). \tag{2.300}$$

2.6.4 Connecting the Position and Momentum Representations

Let us now study how to establish a connection between the position and the momentum representations. By analogy with the foregoing study, when changing from the $\{\mid \vec{r}\rangle\}$ basis to the $\{\mid \vec{p}\rangle\}$ basis, we encounter the *transformation* function $\langle\vec{r} \mid \vec{p}\rangle$.

To find the expression for the transformation function $\langle\vec{r} \mid \vec{p}\rangle$, let us establish a connection between the position and momentum representations of the state vector $\mid \psi\rangle$:

$$\langle\vec{r} \mid \psi\rangle = \langle\vec{r} \mid \left(\int d^3p \mid \vec{p}\rangle\langle\vec{p} \mid\right) \mid \psi\rangle = \int d^3p \langle\vec{r} \mid \vec{p}\rangle\Psi(\vec{p}); \tag{2.301}$$

that is,

$$\psi(\vec{r}) = \int d^3p\, \langle\vec{r} \mid \vec{p}\rangle\Psi(\vec{p}). \tag{2.302}$$

Similarly, we can write

$$\Psi(\vec{p}) = \langle\vec{p} \mid \psi\rangle = \langle\vec{p} \mid \int d^3r \mid \vec{r}\rangle\langle\vec{r} \mid \psi\rangle = \int d^3r \langle\vec{p} \mid \vec{r}\rangle\psi(\vec{r}). \tag{2.303}$$

The last two relations imply that $\psi(\vec{r})$ and $\Psi(\vec{p})$ are to be viewed as Fourier transforms of each other. In quantum mechanics the Fourier transform of a function $f(\vec{r})$ is given by

$$f(\vec{r}) = \frac{1}{(2\pi\hbar)^{3/2}} \int d^3p\, e^{i\vec{p}\cdot\vec{r}/\hbar} g(\vec{p}); \tag{2.304}$$

notice the presence of Planck's constant. Hence the function $\langle \vec{r} \mid \vec{p} \rangle$ is given by

$$\boxed{\langle \vec{r} \mid \vec{p} \rangle = \frac{1}{(2\pi\hbar)^{3/2}} e^{i\vec{p}\cdot\vec{r}/\hbar}.} \tag{2.305}$$

This function transforms from the momentum to the position representation. The function corresponding to the inverse transformation, $\langle \vec{p} \mid \vec{r} \rangle$, is given by

$$\boxed{\langle \vec{p} \mid \vec{r} \rangle = \langle \vec{r} \mid \vec{p} \rangle^* = \frac{1}{(2\pi\hbar)^{3/2}} e^{-i\vec{p}\cdot\vec{r}/\hbar}.} \tag{2.306}$$

The quantity $|\langle \vec{r} \mid \vec{p} \rangle|^2$ represents the probability density of finding the particle in a region around $\vec{r}$ where its momentum is equal to $\vec{p}$.

Remark

If the position wave function

$$\psi(\vec{r}) = \frac{1}{(2\pi\hbar)^{3/2}} \int d^3p\, e^{i\vec{p}\cdot\vec{r}/\hbar} \Psi(\vec{p}) \tag{2.307}$$

is normalized (i.e., $\int d^3r\, \psi(\vec{r})\psi^*(\vec{r}) = 1$), its Fourier transform

$$\Psi(\vec{p}) = \frac{1}{(2\pi\hbar)^{3/2}} \int d^3r\, e^{-i\vec{p}\cdot\vec{r}/\hbar} \psi(\vec{r}) \tag{2.308}$$

must also be normalized, since

$$\begin{aligned} \int d^3p\, \Psi^*(\vec{p})\Psi(\vec{p}) &= \int d^3p \Psi^*(\vec{p}) \left[\frac{1}{(2\pi\hbar)^{3/2}} \int d^3r\, e^{-i\vec{p}\cdot\vec{r}/\hbar} \psi(\vec{r}) \right] \\ &= \int d^3r\, \psi(\vec{r}) \left[\frac{1}{(2\pi\hbar)^{3/2}} \int d^3p\, \Psi^*(\vec{p}) e^{-i\vec{p}\cdot\vec{r}/\hbar} \right] \\ &= \int d^3r\, \psi(\vec{r})\psi^*(\vec{r}) \\ &= 1. \end{aligned} \tag{2.309}$$

This result is known as *Parseval's theorem*.

2.6.4.1 Momentum Operator in the Position Representation

To determine the form of the momentum operator $\hat{\vec{P}}$ in the position representation, let us calculate $\langle \vec{r} \mid \hat{\vec{P}} \mid \psi \rangle$:

$$\begin{aligned} \langle \vec{r} \mid \hat{\vec{P}} \mid \psi \rangle &= \int \langle \vec{r} \mid \hat{\vec{P}} \mid \vec{p} \rangle \langle \vec{p} \mid \psi \rangle d^3p = \int \vec{p} \langle \vec{r} \mid \vec{p} \rangle \langle \vec{p} \mid \psi \rangle d^3p \\ &= \frac{1}{(2\pi\hbar)^{3/2}} \int \vec{p}\, e^{i\vec{p}\cdot\vec{r}/\hbar} \Psi(\vec{p}) d^3p, \end{aligned} \tag{2.310}$$

where we have used the relation $\int \mid \vec{p}\rangle\langle\vec{p} \mid d^3p = \hat{I}$ along with Eq. (2.305). Now, since $\vec{p}\, e^{i\vec{p}\cdot\vec{r}/\hbar} = -i\hbar\vec{\nabla}e^{i\vec{p}\cdot\vec{r}/\hbar}$, and using Eq. (2.305) again, we can rewrite (2.310) as

$$\begin{aligned}\langle\vec{r} \mid \hat{\vec{P}} \mid \psi\rangle &= -i\hbar\vec{\nabla}\left(\frac{1}{(2\pi\hbar)^{3/2}}\int e^{i\vec{p}\cdot\vec{r}/\hbar}\Psi(\vec{p})d^3p\right)\\ &= -i\hbar\vec{\nabla}\left(\int\langle\vec{r} \mid \vec{p}\rangle\langle\vec{p} \mid \psi\rangle d^3p\right)\\ &= -i\hbar\vec{\nabla}\langle\vec{r} \mid \psi\rangle. \end{aligned} \tag{2.311}$$

Thus, $\hat{\vec{P}}$ is given in the position representation by

$$\boxed{\hat{\vec{P}} = -i\hbar\vec{\nabla}.} \tag{2.312}$$

Its Cartesian components are

$$\boxed{\hat{P}_x = -i\hbar\frac{\partial}{\partial x}, \qquad \hat{P}_y = -i\hbar\frac{\partial}{\partial y}, \qquad \hat{P}_z = -i\hbar\frac{\partial}{\partial z}.} \tag{2.313}$$

Note that the form of the momentum operator (2.312) can be derived by simply applying the gradient operator $\vec{\nabla}$ on a *plane* wave function $\psi(\vec{r},t) = Ae^{i(\vec{p}\cdot\vec{r}-Et)/\hbar}$:

$$-i\hbar\vec{\nabla}\psi(\vec{r},t) = \vec{p}\,\psi(\vec{r},t) = \hat{\vec{P}}\psi(\vec{r},t). \tag{2.314}$$

It is easy to verify that $\hat{\vec{P}}$ is Hermitian (see equation (2.378)).

Now, since $\hat{\vec{P}} = -i\hbar\vec{\nabla}$, we can write the Hamiltonian operator $\hat{H} = \hat{\vec{P}}^{\,2}/(2m) + \hat{V}$ in the position representation as follows:

$$\boxed{\hat{H} = -\frac{\hbar^2}{2m}\nabla^2 + \hat{V}(\vec{r}) = -\frac{\hbar^2}{2m}\left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2}\right) + \hat{V}(\vec{r}),} \tag{2.315}$$

where ∇^2 is the Laplacian operator; it is given in Cartesian coordinates by $\nabla^2 = \partial^2/\partial x^2 + \partial^2/\partial y^2 + \partial^2/\partial z^2$.

2.6.4.2 Position Operator in the Momentum Representation

The form of the position operator $\hat{\vec{R}}$ in the momentum representation can be easily inferred from the representation of $\hat{\vec{P}}$ in the position space. In momentum space the position operator can be written as follows:

$$\hat{R}_j = i\hbar\frac{\partial}{\partial p_j} \qquad (j = x, y, z) \tag{2.316}$$

or

$$\boxed{\hat{X} = i\hbar\frac{\partial}{\partial p_x}, \qquad \hat{Y} = i\hbar\frac{\partial}{\partial p_y}, \qquad \hat{Z} = i\hbar\frac{\partial}{\partial p_z}.} \tag{2.317}$$

2.6.4.3 Important Commutation Relations

Let us now calculate the commutator $[\hat{R}_j, \hat{P}_k]$ in the position representation. As the separate actions of $\hat{X}\hat{P}_x$ and $\hat{P}_x\hat{X}$ on the wave function $\psi(\vec{r})$ are given by

$$\hat{X}\hat{P}_x\psi(\vec{r}) = -i\hbar x\frac{\partial\psi(\vec{r})}{\partial x}, \tag{2.318}$$

$$\hat{P}_x\hat{X}\psi(\vec{r}) = -i\hbar\frac{\partial}{\partial x}(x\psi(\vec{r})) = -i\hbar\psi(\vec{r}) - i\hbar x\frac{\partial\psi(\vec{r})}{\partial x}, \tag{2.319}$$

we have

$$\begin{aligned}[\hat{X}, \hat{P}_x]\psi(\vec{r}) &= \hat{X}\hat{P}_x\psi(\vec{r}) - \hat{P}_x\hat{X}\psi(\vec{r}) = -i\hbar x\frac{\partial\psi(\vec{r})}{\partial x} + i\hbar\psi(\vec{r}) + i\hbar x\frac{\partial\psi(\vec{r})}{\partial x} \\ &= i\hbar\psi(\vec{r})\end{aligned} \tag{2.320}$$

or

$$[\hat{X}, \hat{P}_x] = i\hbar. \tag{2.321}$$

Similar relations can be derived at once for the y and the z components:

$$\boxed{[\hat{X}, \hat{P}_x] = i\hbar, \qquad [\hat{Y}, \hat{P}_Y] = i\hbar, \qquad [\hat{Z}, \hat{P}_Z] = i\hbar.} \tag{2.322}$$

We can verify that

$$[\hat{X}, \hat{P}_y] = [\hat{X}, \hat{P}_z] = [\hat{Y}, \hat{P}_x] = [\hat{Y}, \hat{P}_z] = [\hat{Z}, \hat{P}_x] = [\hat{Z}, \hat{P}_y] = 0, \tag{2.323}$$

since the x, y, z degrees of freedom are independent; the previous two relations can be grouped into

$$\boxed{[\hat{R}_j, \hat{P}_k] = i\hbar\delta_{jk}, \qquad [\hat{R}_j, \hat{R}_k] = 0, \qquad [\hat{P}_j, \hat{P}_k] = 0 \qquad (j,\ k = x, y, z).} \tag{2.324}$$

These relations are often called the *canonical commutation relations*.

Now, from (2.321) we can show that (for the proof see Problem 2.8 on page 139)

$$\boxed{[\hat{X}^n, \hat{P}_x] = i\hbar n\hat{X}^{n-1}, \qquad [\hat{X}, \hat{P}_x^n] = i\hbar n\hat{P}_x^{n-1}.} \tag{2.325}$$

Following the same procedure that led to (2.320), we can obtain a more general commutation relation of $\hat{P}_x$ with an arbitrary function $f(\hat{X})$:

$$\boxed{[f(\hat{X}), \hat{P}_x] = i\hbar\frac{df(\hat{X})}{d\hat{X}} \quad \Longrightarrow \quad \left[\hat{\vec{P}},\ F(\hat{\vec{R}})\right] = -i\hbar\vec{\nabla}F(\hat{\vec{R}})),} \tag{2.326}$$

where F is a function of the operator $\hat{\vec{R}}$.

The explicit form of operators thus depends on the representation adopted. We have seen, however, that the *commutation relations for operators are representation independent*. In particular, the commutator $[\hat{R}_j, \hat{P}_k]$ is given by $i\hbar\delta_{jk}$ in the position and the momentum representations; see the next example.

Example 2.20 (Commutators are representation independent)
Calculate the commutator $[\hat{X}, \hat{P}]$ in the momentum representation and verify that it is equal to $i\hbar$.

Solution
As the operator $\hat{X}$ is given in the momentum representation by $\hat{X} = i\hbar\partial/\partial p$, we have

$$\begin{aligned}[\hat{X}, \hat{P}]\psi(p) &= \hat{X}\hat{P}\psi(p) - \hat{P}\hat{X}\psi(p) = i\hbar\frac{\partial}{\partial p}\left(p\psi(p)\right) - i\hbar p\frac{\partial\psi(p)}{\partial p} \\ &= i\hbar\psi(p) + i\hbar p\frac{\partial\psi(p)}{\partial p} - i\hbar p\frac{\partial\psi(p)}{\partial p} = i\hbar\psi(p). \end{aligned} \tag{2.327}$$

Thus, the commutator $[\hat{X}, \hat{P}]$ is given in the *momentum representation* by

$$[\hat{X}, \hat{P}] = \left[i\hbar\frac{\partial}{\partial p},\ \hat{P}\right] = i\hbar. \tag{2.328}$$

The commutator $[\hat{X}, \hat{P}]$ was also shown to be equal to $i\hbar$ in the *position representation* (see equation (2.321):

$$[\hat{X}, \hat{P}] = -\left[\hat{X},\ i\hbar\frac{\partial}{\partial p_x}\right] = i\hbar. \tag{2.329}$$

2.6.5 Parity Operator

The *space reflection* about the origin of the coordinate system is called an *inversion* or a *parity* operation. This transformation is *discrete*. The parity operator $\hat{\mathcal{P}}$ is defined by its action on the kets $|\,\vec{r}\rangle$ of the position space:

$$\hat{\mathcal{P}}\,|\,\vec{r}\rangle = |\,-\vec{r}\rangle, \qquad \langle\vec{r}\,|\,\hat{\mathcal{P}}^\dagger = \langle -\vec{r}\,|, \tag{2.330}$$

such that

$$\hat{\mathcal{P}}\psi(\vec{r}) = \psi(-\vec{r}). \tag{2.331}$$

The parity operator is Hermitian, $\hat{\mathcal{P}}^\dagger = \hat{\mathcal{P}}$, since

$$\begin{aligned}\int d^3r\ \phi^*(\vec{r})\left[\hat{\mathcal{P}}\psi(\vec{r})\right] &= \int d^3r\phi^*(\vec{r})\psi(-\vec{r}) = \int d^3r\ \phi^*(-\vec{r})\psi(\vec{r}) \\ &= \int d^3r\ \left[\hat{\mathcal{P}}\phi(\vec{r})\right]^*\psi(\vec{r}). \end{aligned} \tag{2.332}$$

From the definition (2.331), we have

$$\hat{\mathcal{P}}^2\psi(\vec{r}) = \hat{\mathcal{P}}\psi(-\vec{r}) = \psi(\vec{r}); \tag{2.333}$$

hence $\hat{\mathcal{P}}^2$ is equal to the unity operator:

$$\hat{\mathcal{P}}^2 = \hat{I} \qquad \text{or} \qquad \hat{\mathcal{P}} = \hat{\mathcal{P}}^{-1}. \tag{2.334}$$

The parity operator is therefore *unitary*, since its Hermitian adjoint is equal to its inverse:

$$\hat{\mathcal{P}}^{\dagger} = \hat{\mathcal{P}}^{-1}. \tag{2.335}$$

Now, since $\hat{\mathcal{P}}^2 = \hat{I}$, the eigenvalues of $\hat{\mathcal{P}}$ are $+1$ or -1 with the corresponding eigenstates

$$\hat{\mathcal{P}}\psi_+(\vec{r}) = \psi_+(-\vec{r}) = \psi_+(\vec{r}), \qquad \hat{\mathcal{P}}\psi_-(\vec{r}) = \psi_-(-\vec{r}) = -\psi_-(\vec{r}). \tag{2.336}$$

The eigenstate $\mid \psi_+\rangle$ is said to be *even* and $\mid \psi_-\rangle$ is *odd*. Therefore, the eigenfunctions of the parity operator have *definite parity*: they are either even or odd.

Since $\mid \psi_+\rangle$ and $\mid \psi_-\rangle$ are joint eigenstates of the same Hermitian operator $\hat{\mathcal{P}}$ but with different eigenvalues, these eigenstates must be orthogonal:

$$\langle \psi_+ \mid \psi_- \rangle = \int d^3r \; \psi_+^*(-\vec{r})\psi_-(-\vec{r}) \equiv -\int d^3r \; \psi_+^*(\vec{r})\psi_-(\vec{r}) = -\langle \psi_+ \mid \psi_- \rangle; \tag{2.337}$$

hence $\langle \psi_+ \mid \psi_- \rangle$ is zero. The states $\mid \psi_+\rangle$ and $\mid \psi_-\rangle$ form a complete set since any function can be written as $\psi(\vec{r}) = \psi_+(\vec{r}) + \psi_-(\vec{r})$, which leads to

$$\psi_+(\vec{r}) = \frac{1}{2}\left[\psi(\vec{r}) + \psi(-\vec{r})\right], \qquad \psi_-(\vec{r}) = \frac{1}{2}\left[\psi(\vec{r}) - \psi(-\vec{r})\right]. \tag{2.338}$$

Since $\hat{\mathcal{P}}^2 = I$ we have

$$\hat{\mathcal{P}}^n = \begin{cases} \hat{\mathcal{P}} & \text{when } n \text{ is odd,} \\ \hat{I} & \text{when } n \text{ is even.} \end{cases} \tag{2.339}$$

Even and odd operators

An operator $\hat{A}$ is said to be *even* if it obeys the condition

$$\hat{\mathcal{P}}\hat{A}\hat{\mathcal{P}} = \hat{A} \tag{2.340}$$

and an operator $\hat{B}$ is *odd* if

$$\hat{\mathcal{P}}\hat{B}\hat{\mathcal{P}} = -\hat{B}. \tag{2.341}$$

We can easily verify that even operators commute with the parity operator $\hat{\mathcal{P}}$ and that odd operators anticommute with $\hat{\mathcal{P}}$:

$$\hat{A}\hat{\mathcal{P}} = (\hat{\mathcal{P}}\hat{A}\hat{\mathcal{P}})\hat{\mathcal{P}} = \hat{\mathcal{P}}\hat{A}\hat{\mathcal{P}}^2 = \hat{\mathcal{P}}\hat{A}, \tag{2.342}$$

$$\hat{B}\hat{\mathcal{P}} = -(\hat{\mathcal{P}}\hat{B}\hat{\mathcal{P}})\hat{\mathcal{P}} = -\hat{\mathcal{P}}\hat{B}\hat{\mathcal{P}}^2 = -\hat{\mathcal{P}}\hat{B}. \tag{2.343}$$

The fact that even operators commute with the parity operator has very useful consequences. Let us examine the following two important cases depending on whether an even operator has nondegenerate or degenerate eigenvalues:

- If an even operator is Hermitian and none of its eigenvalues is degenerate, then this operator has the same eigenvectors as those of the parity operator. And since the eigenvectors of the parity operator are either even or odd, the eigenvectors of an even, Hermitian, and nondegenerate operator must also be either even or odd; they are said to have a *definite parity*. This property will have useful applications when we solve the Schrödinger equation for even Hamiltonians.